How Much Targeted Training Does It Take to Top a Multi-Task Leaderboard? Minimum Investment Under Different Aggregation Rules

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Abstract

A leaderboard turns per-task scores into one rank, so a developer can try to lift a chosen model by training on data related to individual sub-datasets. How much such targeted training is needed to put the model first, and how does that depend on the aggregation rule? We define the investment as the minimum raise of the target's task scores, within a per-task cap, that makes it strictly first, and measure it in points, tasks touched and largest single-task jump. We solve it exactly for score rules (mean, median, minimum, power means); for rank rules (Borda, Condorcet winner, plurality; Copeland on HELM) an integer program under strict comparisons decides feasibility and tasks touched exactly and returns the points as a witness cost within $n\varepsilon$ of the infimum. We prove closed forms for the mean, minimum and odd median, and apply it to every model of the Open LLM Leaderboard (OLLB; 4495 targets, 6 tasks) and of HELM MMLU (78 targets, 57 subjects). On OLLB the rank rules are solved only for the 49 best non-leading targets (mean-ranks 2–50), so comparisons involving them use those targets alone. With caps taken from observed fine-tuning gains, first place is reachable for 12% of OLLB targets under the mean (bootstrap interval 11–17%; every target up to rank 394 is feasible; the largest feasible rank is 901) and for 19% of HELM targets (15–42%; up to rank 16); with a 100-point cap every target can reach it. The rules do not yield one ordering. In points, on targets feasible with positive cost under both rules, the median needs a median of 0.39 of the mean's total on OLLB (0.07 on HELM) and Borda 0.27 (0.51), but not at the very top: the OLLB targets of mean-rank 2–4 need more points under the median than under the mean. In reach, the OLLB median makes fewer targets feasible than the mean (10% against 12%, all-feasible prefix 90 against 394) although its largest feasible rank is larger (1207 against 901; the two shares' intervals overlap), while on HELM its point estimate is higher (33% against 19%, intervals overlapping). Conclusions are conditional on the cap, which on HELM is transferred from OLLB, and hold for fixed rivals and fixed score tables.

1 Introduction

Multi-task benchmarks rank models by aggregating per-task scores into one number, and the resulting rank influences which models are adopted, published and funded. A developer therefore has an incentive to improve a chosen target model specifically on the benchmark's sub-datasets, for instance by training on data related to them. How much such targeted improvement is needed to reach first place, and how that amount depends on the aggregation rule, is the question we study on public leaderboard data.

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Our approach is deliberately simple. We take the score table of a public leaderboard as fixed, let the target raise its own scores task by task up to a cap, and compute the *minimum* raise that makes it strictly first under a given aggregation rule. The cap is the central modelling assumption: it stands for how far training on related data can move one score. We therefore report results under data-grounded caps and an oracle cap, and report infeasibility (no allowed raise makes the target first) as an outcome rather than a missing value.

Contributions. (i) A formulation of the minimum targeted investment with three cost measures (L_1, K, L_∞) and an explicit family of caps (Section 3). (ii) Exact solvers for score rules (mean, median, minimum, geometric, harmonic and squared power means) and integer programs for rank rules (Borda, Condorcet winner, plurality, Copeland) that decide feasibility and K exactly and return an L_1 witness cost within $n\varepsilon$ of the infimum, all tested against brute force, with closed forms and proofs for the mean, minimum and odd-n median (Section 3.4, Appendix A). (iii) A study of every model on two public leaderboards, with bootstrap and evaluation-noise ranges, paired same-target rule comparisons and a feasibility frontier (Sections 4–5); the rank rules on OLLB cover only the 49 best non-leading targets, as stated wherever they appear.

Findings. Under caps grounded in observed fine-tuning gains, first place is out of reach for most models under every rule studied: the feasible fraction under the mean is 12% on OLLB and 19% on HELM. The rules differ in two ways that do not combine into one ordering. First, *cost on the same targets*: on targets feasible with positive cost under both rules, order-statistic and rank rules need a fraction of the mean's total points (median-of-ratios on OLLB: median 0.39, minimum 0.11, Borda 0.27, Condorcet 0.52; the Borda, Condorcet and plurality ratios use the 36–48 best targets only), with exceptions at the very top, where the OLLB median costs more than the mean for the targets of mean-rank 2–4. Second, *reach*: the OLLB median is feasible for fewer targets than the mean (10% against 12%) and has a shorter all-feasible prefix (90 against 394) but a larger largest feasible rank (1207 against 901); on HELM the median's point estimate is 33% against 19% for the mean, with overlapping bootstrap intervals. Of three hypotheses fixed beforehand, one holds, one holds in part and one is not supported for the median rule (Section 5). All conclusions are conditional on the cap and on the other models being fixed; Section 6 lists where they fail.

2 Related work

Benchmarks and leaderboards. MMLU [Hendrycks et al., 2020] aggregates 57 subjects, and HELM [Liang et al., 2022] releases per-scenario scores and raw model outputs for many models; BIG-Bench Hard [Suzgun et al., 2022] selects hard tasks from BIG-Bench. These provide the multi-task score matrices that we analyse.

Fragility of rankings. Dehghani et al. [2021] show that the relative order of methods can change with the choice of benchmark tasks (the "benchmark lottery"). Singh et al. [2025] document how private testing and unequal data access distort the Chatbot Arena leaderboard. Both concern what is measured and disclosed; we instead hold the leaderboard fixed and ask what a single developer must change for one model to win.

Aggregation and social choice. Lanctot et al. [2023] treat each task as a voter and use voting theory to build cross-task evaluation; the Handbook of Computational Social Choice [Brandt et al., 2016] covers the rules (Borda, Condorcet-consistent rules, plurality) we reuse. Our rules are the same families, but our question is the cost of changing the winner, not the choice of rule.

Training on the test task and contamination. Zhou et al. [2023] show that benchmark leakage into training can raise scores sharply, and Yang et al. [2023] show that rephrased test samples evade common decontamination. Chen et al. [2026] analyse, in a game-theoretic model, the incentives that benchmarks create for strategic post-training. Our analysis is empirical and combinatorial: given a fixed score table, we compute the exact minimum targeted improvement per rule, under explicit assumptions about how much any one task can be improved.

Positioning. Closest in question is Chen et al. [2026] (we read the full text of arXiv v1). Their developers choose a scalar amount of benchmark-specific effort that a post-effort score maps to one

benchmark score per model; a multi-dimensional capability with a linear benchmark weight vector is mentioned as an extension, but the rule that aggregates tasks is not a variable of their analysis. They prove that current protocols induce games without a Nash equilibrium and that tune-before-test induces a unique equilibrium ranking by latent quality, and in a case study of one benchmark they measure the training steps needed to change a ranking under tune-before-test. We take the opposite stance: no game and no equilibrium, a fixed table, and the exact cost for one developer under each of several aggregation rules, in score points under per-task caps rather than training steps. Colombo et al. [2022] propose a social-choice-based aggregation for NLP benchmarks and show it changes conclusions relative to the mean; they also corrupt the scores of some tasks for all systems at random and find rank-based aggregation more stable than the mean in ranking error. We do not propose a rule, and our perturbation is different: one target raises its own scores, and we compute the minimum cost of becoming first under each rule. The computational-social-choice literature studies the analogous *bribery* problem, the cost of changing votes so that a chosen candidate wins under a voting rule [Faliszewski et al., 2009, Brandt et al., 2016]. Our setting differs in that the manipulator changes its own scores on tasks (the "voters") and pays in score points under per-task caps, which gives closed forms for score rules and integer programs for rank rules.

3 Problem setup

3.1 Leaderboard, target and investment

A leaderboard is a score matrix $X \in [0, 100]^{M \times n}$ of M models on n tasks. Fix a target model i with score vector $x = X_{i\cdot}$; all other rows stay fixed. In targeted training the developer can raise the target's score on task t to any $y_t \in [x_t, c_t]$, where c_t is a *cap* (Section 3.2). An aggregation rule f maps the score vectors to a ranking; the target is *first* if it is strictly ahead under f. For a score rule we require a margin $\varepsilon = 10^{-3}$ points, $f(y) \geq \tau + \varepsilon$ with τ the best rival's value. A rank rule uses plain strict comparisons and no margin: a pairwise comparison of two scores is a win (strictly higher), a loss, or a tie worth one half, and under Borda, plurality and Copeland the target is first if its total is strictly the largest, under the Condorcet requirement if it beats every rival on more than half of the tasks; there ε only sets the granularity of the witness we return. For score rules with $\varepsilon > 0$ the minima below are attained, and the infimum over strict wins is the limit $\varepsilon \to 0$; for rank rules the infimum over strict wins need not be attained (it can be attained at a tie), so the solver returns an explicit witness (Section 3.4). Evaluation noise of the scores is far larger than ε (Section 5), so the score margin defines an exact optimisation problem, not a win that would survive resampling; for score rules, margins of 1 and 2 points are reported as variants. We report the investment in three ways:

$$L_1 = \sum_t (y_t - x_t), \qquad K = \#\{t : y_t > x_t\}, \qquad L_\infty = \max_t (y_t - x_t),$$

the total points, the number of tasks touched, and the largest single-task jump. The minimum of each is taken separately, so the three need not come from one raise. L_∞ is the smallest d for which a raise of every task by at most d (a "general improvement") suffices. For comparison across rules and datasets we also report K/n and L_1/n.

3.2 Caps: how much can one task be improved?

Without a bound on y_t the question is trivial (put every task at 100). The cap c_t is therefore the central modelling assumption, and we never collapse it to a single choice. The primary caps are data-grounded: on OLLB `emp99`, $c_t = \min(100, x_t + g_t^{99})$ where g_t^{99} is the 99th percentile of the observed gains (fine-tune minus its declared base model, 540 pairs) on task t; on HELM, which has no such pairs, `emp99T`, the OLLB MMLU-PRO value $g^{99}_{\text{MMLU-PRO}}$ applied to every subject (a transfer assumption). Both inherit a selection bias, since only published fine-tunes appear, they measure gains attained by the community and not those a determined developer could attain, and the 99th percentile of 540 pairs rests on about five tail points per task. The oracle cap `ceil`, $c_t = 100$, gives a lower bound on any required investment. Further caps (95th percentiles, the best rival's score, and $+5/+10/+20$ points on every task) are reported in the appendix. When no $y \leq c$ makes the target first, the case is *infeasible*; infeasibility is reported as a rate and never coded as a large investment, and all investment statistics are conditional on feasibility.

3.3 Aggregation rules

Score rules apply a function f to the score vector of each model: the *mean* (the rule of both leaderboards we use), the *median*, and the *minimum* (worst-task score). Rank rules use only the order of models on each task: *Borda* (sum over tasks of the number of models beaten, ties counting one half; equivalent to mean rank), the *Condorcet-winner* requirement (the target beats every rival on a strict majority of tasks, ties counting one half), *Copeland* (a model scores 1 for each rival it beats by that pairwise count, one half for a tie; first means the strictly largest score) and *plurality* (each task gives one point, split equally among the models tied for its highest score; first means the strictly largest total). Being a Condorcet winner is sufficient for being first under any Condorcet-consistent rule such as Copeland or Schulze, so its investment upper-bounds theirs. Geometric, harmonic and power means and the trimmed mean are in the appendix.

3.4 Exact minimum-investment solvers

For a score rule let $\tau = \max_{j \neq i} f(X_{j\cdot})$ be the level of the best rival; the target is first iff $f(y) \geq \tau + \varepsilon$. For the mean, the minimum and the median the minimum investment has a closed form (Propositions 1–3, proved in Appendix A); for power means and the trimmed mean we solve a convex program or enumerate exactly. For rank rules the outcome depends only on the order of scores, so we use plain strict comparisons and no margin. The integer program (HiGHS, rival constraints added lazily) lets each raised task end at a rival's score (a tie) or at most ε above it, and returns that explicit witness. By construction and by tests against brute force on fractional, tied and cap-boundary instances, feasibility and K are exact for continuous raises, and the true infimum of L_1 lies within $n\varepsilon$ of the witness cost (Appendix C). A target already ahead of every rival under a score rule is counted first at zero cost, whatever the size of its lead. All solvers are tested against brute force on small instances. For each target we report L_1 and K, each minimised separately; L_∞ is computed but is not part of the headline results.

Write $b = \tau + \varepsilon$ for the level a score rule must reach and $h_t = c_t - x_t$ for the headroom of task t. They concern targets not already ahead ($f(x) \leq \tau$); a target with $f(x) > \tau$ costs zero by the convention above.

Proposition 1 (Mean)**.** *Let f be the mean and $R = n\,b - \sum_t x_t$ the required total. If $R \leq 0$ the target is already first and every cost is zero. Otherwise it can become first iff $\sum_t h_t \geq R$; the minimum L_1 is R whatever its spread; the minimum K is the least k for which the k largest h_t sum to at least R; the minimum L_∞ is the least d with $\sum_t \min(d, h_t) \geq R$, which equals R/n when no cap binds. The capped uniform raise $y_t = \min(x_t + d, c_t)$ with this d attains the minimum L_1 and the minimum L_∞.*

Proposition 2 (Minimum)**.** *Let $f = \min_t y_t$ and $D = \{t : x_t < b\}$. The target can become first iff $c_t \geq b$ for all $t \in D$. Then $L_1 = \sum_{t \in D}(b - x_t)$ and $K = |D|$: every deficient task must be raised, and $L_\infty = \max_{t \in D}(b - x_t)$ (zero if D is empty).*

Proposition 3 (Median, n odd)**.** *Let $m = (n+1)/2$, $a = \#\{t : x_t \geq b\}$ and, for $x_t < b$, $d_t = b - x_t$ if $c_t \geq b$ and $d_t = \infty$ otherwise. The target is first iff at least m tasks end at or above b. If $a \geq m$ no raise is needed; otherwise the minimum L_1 is the sum of the $m - a$ smallest finite d_t and the minimum K is $m - a$, and the target can be made first iff $m - a$ finite d_t exist.*

These are finite-margin statements: the minima are attained at $b = \tau + \varepsilon$. For the strict-win infimum let $\varepsilon \to 0$, which replaces b by τ and every "$\geq b$" by "$> \tau$" in the feasibility conditions; the infimum of L_1 is then $R_0 = n\tau - \sum_t x_t$ for the mean and $\sum_{t \in D_0}(\tau - x_t)$ for the minimum, but for a target that is not already ahead it is not attained (a strict win needs a strictly positive margin). Near ties between a rival's level and a task score the finite-margin K can differ from the infimum's K by the tasks within ε of τ; the code uses $\varepsilon = 10^{-3}$ points throughout. For even n the median is the average of the two middle order statistics and we only have the computed solution. We verify Propositions 1–3 numerically against the solvers (Section 4); no theorem is claimed for the rank rules, which follow the strict model above (Appendix C).

4 Experimental setup

Data. We use two public leaderboards, each at a fixed, recorded version (checksums in the supplementary reproduction notes). *HELM MMLU*, release v1.13.0 (summary dated 2025-01-10): the

exact-match score of 79 models on each of the 57 MMLU subjects (79×57, no missing cell). *Open LLM Leaderboard* (OLLB) `contents`, dataset commit `9c09a7c` (2025-03-20): 4576 rows, of which 4497 have distinct model names after we keep the best-scoring variant of each duplicated name, and 4496 after removing one flagged model; each model has six normalised task scores (IFEval, BBH, MATH Lvl 5, GPQA, MUSR, MMLU-PRO), so $n = 6$. Merged models are kept in the main analysis and dropped in a robustness slice. The mean we use equals the sites' own aggregate (OLLB `Average`, HELM "MMLU All Subjects") to within 6×10^{-14} points on every model. Neither source declares a licence for the score data; we redistribute only the derived score tables.

Targets, caps and what is reported. Every model other than the current leader is a target in turn, with all others fixed (4495 targets on OLLB, 78 on HELM). On OLLB the primary cap is `emp99` (per-task 99th percentile of the gain of a fine-tune over its declared base model, estimated from 540 such pairs in the same leaderboard; this is about five observations in each task's upper tail; its sampling error is quantified below by a bootstrap over the pairs). HELM has no base/fine-tune pairs, so its primary cap, `emp99T`, applies the OLLB MMLU-PRO 99th-percentile gain to all 57 subjects; this is a transfer assumption, not a measurement on HELM. The oracle cap `ceil` and the further caps of Appendix B are reported beside it. All investment statistics are medians over feasible targets, shown together with the feasible fraction. Rules are compared on the same targets only (paired comparisons, with the number of targets in both feasible sets). OLLB rank rules are integer programs that at $M = 4496$ models we run only for the 50 best targets under the mean (49 non-leading targets; mean-ranks 2–50, a 120 s limit per instance, time-outs reported as censored, never as infeasible), plus a variant on the 500 best models (Appendix B); Copeland is solved on HELM and on the 500-best OLLB slice.

5 Results

All statements below are about targets that can be made first within the cap; the feasible fraction is shown beside every cost. A rule is called cheaper than another only on targets feasible under both, and any OLLB statement that involves Borda, Condorcet or plurality refers to the 49 best non-leading targets.

The closed forms are exact (Q1). Propositions 1–3 agree with the exact solvers on every target of both datasets and every cap: the largest deviation in L_1 is below 2×10^{-12} points, the deviation in K is zero, and no feasibility verdict differs (Appendix Table 2). The OLLB median is solver-only because $n = 6$ is even.

Feasibility depends on rank, rule and cap (Q3). Under the data-grounded caps, targeted training cannot make most models first (Figure 1, left). Appendix Table 3 gives three summaries that do not rank the rules the same way. On OLLB under `emp99` the share of feasible targets is 12.1% for the mean (546 of 4495), 10.0% for the median and 6.1% for the minimum; every target is feasible up to rank 394 (mean), 90 (median) and 5 (minimum); the largest feasible rank is 901, 1207 and 2225. A bootstrap over the 540 gain pairs (1000 resamples, 95% percentile intervals; sampling noise of the cap percentile only, not the selection bias of published fine-tunes, and not run for rank rules) gives 12.1% [11.2, 16.6] (mean), 10.0% [8.3, 12.1] (median) and 6.1% [5.0, 8.2] (minimum), and 901 [824, 1201] for the mean's largest feasible rank; the mean and median intervals overlap, the minimum's lies below. The median's point estimate is below the mean's, with a shorter all-feasible prefix, but it reaches isolated targets farther down; the minimum reaches the farthest isolated target but is rarely feasible. On the 49 non-leading targets in the exactly solved set, 49 are feasible under the mean, the median and Borda, 42 under the Condorcet requirement and 36 under plurality, so these two rank rules are harder to reach than the mean; on the 500-best slice (Appendix B) the shares are 95% (mean), 91% (Borda), 79% (median), 56% (plurality), 40% (minimum) and 30% (Condorcet). On HELM under `emp99T` the shares are 19% [15, 42] (mean), 33% [21, 55] (median), 12% [9, 21] (minimum), 21% (Borda), 32% (Condorcet and Copeland) and 58% (plurality), with all-feasible prefixes of 16, 19, 6, 17, 26, 26 and 39; on HELM the point estimates of the median and Condorcet-type rules exceed the mean's, but the median and mean intervals overlap heavily, so we claim no reach ordering on HELM from this cap alone. The ordering is also a function of the cap: on OLLB the minimum's share is 2.3%, 9.9% and 64% under uniform caps of $+5$, $+10$ and $+20$ points, against 0.3%, 2.5% and 18% for the mean, so the minimum is the most reachable rule under these uniform caps and the least

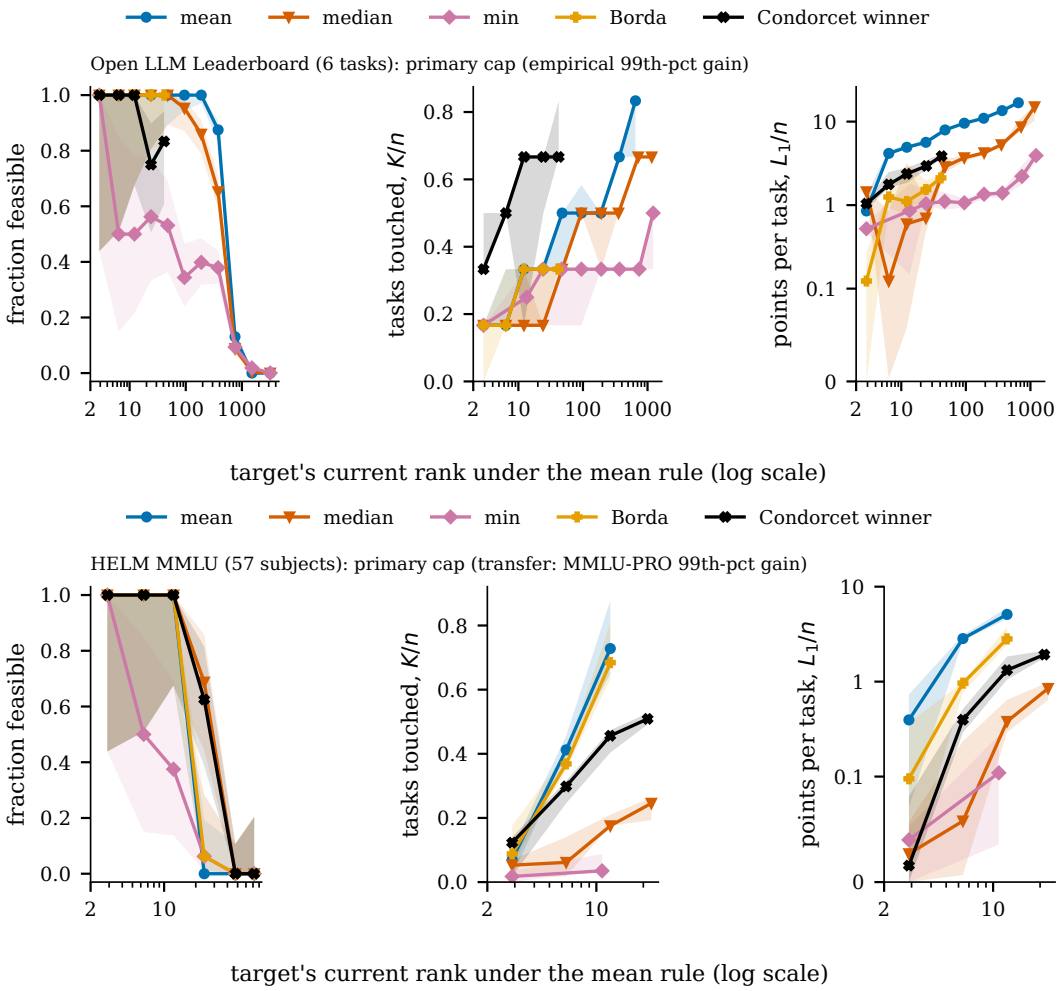


Figure 1: Minimum targeted investment to become first versus the target's current rank under the mean, primary caps. Left: fraction of targets that can reach first at all. Middle and right: tasks touched (K/n) and total points per task (L_1/n), *among feasible targets*. Top: OLLB, $n = 6$, cap `emp99` (4495 targets; Borda and Condorcet only for the 49 best non-leading targets, ranks 2–50). Bottom: HELM MMLU, $n = 57$, cap `emp99T` (78 targets; the cap is transferred from OLLB). Points are bin medians (bins double from rank 2; bins with fewer than 3 targets omitted), bands are 95% bootstrap intervals over targets. The feasible sets differ by rule, so vertical gaps between rules in the middle and right panels compare different targets; same-target comparisons are in Table 1. Generated by `figures/src/fig_rank_rules.py`.

reachable under the per-task empirical caps; on HELM the median's share is at least the mean's under all five finite caps, and the minimum's is below it (Appendix Figure 4). The cap, not the rule, decides feasibility far from the top: with the oracle cap `ceil` every target is feasible (Appendix Figure 3).

Cost on the same targets: typically lower than the mean's, with exceptions at the top (Q2). Table 1 compares each rule with the mean on the targets feasible under both with positive investment under both (we exclude targets with zero cost under either rule from the ratios; targets already first count as feasible in all reach statistics). Over all such OLLB targets the median of the per-target ratio $L_1^{\text{rule}}/L_1^{\text{mean}}$ is 0.39 for the median rule (95% interval 0.38–0.40; 435 targets, cheaper than the mean for 99% of them), 0.11 for the minimum (237 targets, cheaper for all), 0.27 for Borda (48), 0.52 for the Condorcet requirement (42; cheaper for 95%) and 0.66 for plurality (36; cheaper for 78%); the Borda, Condorcet and plurality figures cover the top 50 only. The median and minimum also touch fewer tasks (median ratio of K 0.75 and 0.50), Borda the same number and the Condorcet

requirement twice as many. On HELM the ratios are 0.07 (median rule, 15 targets), 0.05 (minimum, 7), 0.51 (Borda, 15), 0.21 (Condorcet and Copeland, 15) and 0.05 (plurality, 14); only 15 of the 78 targets are feasible under the mean, so the HELM statements rest on at most 15 targets. The reason is structural: the mean must close n times the gap (Proposition 1), order-statistic rules only the cheapest majority or the weakest tasks.

This ordering is a statement about typical targets, not a dominance result. At the top of OLLB it fails: the targets of mean-rank 2, 3 and 4 need 9.55, 8.68 and 8.37 points under the median rule against 4.77, 5.11 and 7.90 under the mean, and the rank-2 target needs 6.70 points under the Condorcet requirement (rank 3 and 4: 5.29, 6.25); only at rank 5 is the median far cheaper (0.70 against 23.86). The median ratio in the 2–5 stratum is 1.38 with an interval that spans 0.03–2.0 over four targets, and the median is cheaper for one of them (Appendix Table 4). On HELM the rank-2 target (DeepSeek v3) needs 5.58 points under Borda against 4.04 under the mean, and is already first under the minimum rule and within 0.001 point of first under the median, Condorcet and Copeland rules. A uniform capped raise of every task needs a median L_1 of 2.4 (median rule), 3.6 (minimum), 2.6 (Borda), 2.4 (Condorcet) and 3.0 (plurality) times the optimum on OLLB and 9.0, 38, 1.9 and 3.0 on HELM (Appendix Table 5); for the mean the two coincide. The OLLB ordering mean $>$ Condorcet $>$ Borda $\gtrsim$ median $\gtrsim$ min in typical cost is therefore not uniform across rank bins; the ratio intervals for ranks 2–10 are wide.

Which of our predictions held. The plan fixed three hypotheses before the full results (H1–H3, Appendix D). H2 (cost rises with rank, feasibility falls as caps tighten) holds on both datasets. H1 (order-statistic and majority rules need *more* tasks but possibly fewer points than the mean) holds only in part: they need fewer points on typical targets (with the exceptions above) but not more tasks, except that the Condorcet requirement touches more tasks than the mean on OLLB (ratio of K about 2); on HELM the median needs fewer tasks than the mean (K ratio 0.27). H3 is examined next.

Number of tasks (Q4). Appendix Figure 6 draws random subsets of $n \in \{3, 6, 12, 24, 48\}$ HELM subjects and, in each subset, takes as the target the model ranked second or fifth *under the rule being studied*, so the target model differs between rules and the curves of different rules mix the effect of the rule with that of the target. For the rank-2 target K/n falls with n under every rule. For the rank-5 target the mean and Borda rules keep touching about 0.4 of the tasks (median K 9 of 24 and 19 of 48 under the mean) while the median, minimum and plurality fall (median rule: $K = 1$ at $n = 3$ to $K = 3$ at $n = 48$); the mean's L_1/n stays between 2.2 and 2.8 while the median rule's falls from 0.8 to 0.04. H3 (that K/n of the median and majority rules does not shrink) is therefore not supported for the median on these own-rule targets. At rank 5 first place is reachable in only 55–90% of the subsets for the minimum and 28% (at $n = 3$) for plurality, so their cost curves are conditional on feasible subsets; the bands are wide (60 draws per n). A common-target run (Appendix Figure 7: the target is the model ranked 2nd or 5th under the *mean* of each subset, one target per draw for all rules) separates the rule from the target. For the rank-5 target the median investment under the mean grows 20-fold from $n = 3$ to $n = 48$ (6.6 to 133 points), against 2.8-fold under the median rule (2.7 to 7.4), 1.8 under the minimum (3.6 to 6.5), 1.7 under plurality (3.5 to 5.9) and 11 under Borda (3.7 to 41), as Proposition 1 predicts for a rule that must close n times the gap; K/n of the median rule falls from 0.33 to 0.10, so H3 is not supported on the common target either. At rank 2 many draws already make the target first under the other rules (at $n = 48$: Borda 18, median 21, minimum 32 of 60 draws), which pulls their median cost towards zero. OLLB ($n = 6$) supports no scaling claim.

Robustness (Q5). Appendix Figure 2 perturbs the leaderboard and recomputes the *optimal* requirement on each draw. Two perturbations are used: independent Gaussian noise on every score with the binomial standard error from the item counts and a bootstrap over tasks (draw counts and the equal-item-count assumption for OLLB are in the figure caption). What the runs measure is the variability of the optimal requirement, not whether a fixed, previously optimised raise would still win (the fixed-policy check is below). Under evaluation noise the 95% range of L_1 under the mean is about 3–3.6 times its median for the OLLB targets of rank 2–3 and shrinks to 0.4–0.8 times for ranks 5–100; on HELM it is 14 times at rank 2, 2.7 at rank 3 and 0.5–0.7 at ranks 5–10. For the median rule on OLLB the ranges are 0.4–3.0 times the median, and the minimum rule is infeasible in 35–100% of the OLLB draws. A target whose gap is within the evaluation noise therefore has no stable requirement: at ranks 2–3 some draws already make it first (lower end of the range 0). The observed requirement lies inside the 2.5–97.5 range in every feasible case shown (caption), which

Table 1: Median investment to become first, by stratum of the target's rank under the mean, primary caps: OLLB (`emp99`, upper block) and HELM (`emp99T`, lower block). *feas.*: targets that can reach first place / targets in the stratum († OLLB Borda and Condorcet: the part of the stratum with rank ≤ 50, hence 20/20 in 31–100). K and L_1: medians over the rule's feasible targets with 95% bootstrap intervals over targets; these are medians over different target sets for different rules and are not a ranking of the rules. L_1/L_1^{mean}: median [interval] of the per-target ratio over targets feasible under the rule and the mean with positive cost under both (same targets; from `paired_*.csv`), and n_{both}: those targets / targets feasible under at least one of the two. "–": the mean itself, fewer than 3 jointly feasible targets, or no group in the analysis (HELM 2–5; see Appendix Table 4). Generated by `figures/src/tables_v2.py`.

rule	feas.	K [95% CI]	L_1 [95% CI]	L_1 / L_1^{mean} [95% CI]	n_{both}
target's rank under the mean: 2–5 (4 targets)					
mean	4/4	1.0 [1.0, 1.0]	6.5 [4.8, 23.9]	–	–
median	4/4	1.0 [1.0, 1.0]	8.5 [0.7, 9.6]	1.38 [0.03, 2.00]	4/4
min	4/4	1.0 [1.0, 2.0]	3.4 [2.8, 14.6]	0.61 [0.35, 0.77]	4/4
Borda†	4/4	1.0 [0.0, 1.0]	1.2 [0.0, 6.2]	0.26 [0.09, 0.35]	3/4
Condorcet†	4/4	2.5 [2.0, 3.0]	6.5 [5.3, 9.7]	0.91 [0.41, 1.41]	4/4
target's rank under the mean: 6–10 (5 targets)					
mean	5/5	2.0 [1.0, 2.0]	27.7 [24.6, 28.5]	–	–
median	5/5	1.0 [0.0, 1.0]	0.3 [0.0, 5.0]	0.02 [0.00, 0.18]	4/5
min	1/5	2.0 [2.0, 2.0]	6.2 [6.2, 6.2]	–	1/5
Borda†	5/5	2.0 [1.0, 2.0]	5.8 [5.0, 8.9]	0.20 [0.18, 0.36]	5/5
Condorcet†	5/5	3.0 [3.0, 4.0]	14.9 [10.0, 16.9]	0.59 [0.40, 0.59]	5/5
target's rank under the mean: 11–30 (20 targets)					
mean	20/20	2.0 [2.0, 2.0]	32.2 [31.8, 34.0]	–	–
median	20/20	1.0 [1.0, 1.0]	4.2 [3.0, 10.0]	0.13 [0.09, 0.29]	20/20
min	12/20	2.0 [1.0, 2.0]	6.7 [4.8, 9.3]	0.21 [0.10, 0.28]	12/20
Borda†	20/20	2.0 [1.0, 2.0]	9.4 [7.7, 12.1]	0.27 [0.23, 0.37]	20/20
Condorcet†	16/20	4.0 [3.0, 4.0]	16.7 [11.7, 20.2]	0.51 [0.37, 0.56]	16/20
target's rank under the mean: 31–100 (70 targets)					
mean	70/70	3.0 [3.0, 3.0]	52.6 [50.9, 53.8]	–	–
median	69/70	2.0 [2.0, 2.0]	19.5 [17.2, 21.6]	0.37 [0.32, 0.40]	69/70
min	30/70	2.0 [1.0, 2.0]	6.3 [5.3, 8.3]	0.14 [0.10, 0.18]	29/70
Borda†	20/20	2.0 [2.0, 2.0]	12.5 [11.5, 13.7]	0.27 [0.25, 0.31]	20/70
Condorcet†	17/20	4.0 [4.0, 5.0]	23.1 [21.9, 24.5]	0.49 [0.47, 0.57]	17/70

rule	feas.	K [95% CI]	L_1 [95% CI]	L_1 / L_1^{mean} [95% CI]	n_{both}
target's rank under the mean: 2–5 (4 targets)					
mean	4/4	5.0 [1.0, 22.0]	32.2 [4.0, 154.0]	–	–
median	4/4	3.5 [1.0, 5.0]	2.4 [0.0, 6.2]	–	–
min	4/4	1.5 [0.0, 3.0]	2.6 [0.0, 7.6]	–	–
Borda	4/4	7.5 [3.0, 20.0]	13.3 [0.5, 43.7]	–	–
Condorcet	4/4	7.5 [1.0, 18.0]	2.8 [0.0, 24.5]	–	–
target's rank under the mean: 6–10 (5 targets)					
mean	5/5	26.0 [23.0, 37.0]	176.8 [156.3, 260.2]	–	–
median	5/5	8.0 [1.0, 10.0]	13.5 [0.4, 16.6]	0.06 [0.00, 0.09]	5/5
min	2/5	1.5 [1.0, 2.0]	2.2 [2.0, 2.4]	–	2/5
Borda	5/5	23.0 [20.0, 37.0]	56.5 [54.0, 152.8]	0.36 [0.32, 0.59]	5/5
Condorcet	5/5	18.0 [14.0, 23.0]	29.3 [16.5, 62.0]	0.19 [0.09, 0.28]	5/5
target's rank under the mean: 11–20 (10 targets)					
mean	6/10	46.0 [40.0, 51.0]	321.2 [278.2, 351.3]	–	–
median	9/10	11.0 [10.0, 12.0]	36.4 [20.3, 37.6]	0.09 [0.07, 0.11]	6/9
min	3/10	3.0 [2.0, 5.0]	10.4 [6.2, 15.5]	–	2/7
Borda	7/10	45.0 [38.0, 48.0]	187.8 [143.1, 230.8]	0.55 [0.51, 0.64]	6/7
Condorcet	10/10	27.5 [26.0, 28.5]	101.1 [77.2, 106.6]	0.28 [0.21, 0.31]	6/10

with ranges this wide does not bound a systematic shift. Holding the optimal raise fixed instead (`results/robust/survival_*.csv`; for 7 OLLB and 6 HELM targets of mean-rank 2–100 and 2–40, 300 and 200 noise draws each, noise independent across cells, the policy not re-optimised; Appendix Table 6), the target is first in a minority of draws at the smallest tested score margin or ordinal witness granularity (10^{-3} points): the median over these targets of that share is 0.43 (OLLB) and 0.38 (HELM) for the mean, 0.21 and 0.05 for the median rule, 0.11 and 0.05 for the minimum and 0.19 (Borda), 0.18 (plurality) and 0.01 (Condorcet) on OLLB. A 1- and 2-point margin lifts the mean rule to 0.80 and 0.98 (OLLB) and 0.96 and 1.00 (HELM), and the OLLB median rule to 0.72 and 0.96, but not the HELM median rule (0.11, 0.28) or the minimum (0.21, 0.45 on OLLB).

6 Discussion and limitations

What the results say. Under caps taken from observed fine-tuning gains, a developer who can lift only the target's own scores, with rivals standing still, tops these leaderboards only from near the top; how near depends on the rule and the criterion. On the same targets the mean rule typically demands the most points, because it must close n times the gap (Proposition 1) whatever the spread, while the median and minimum pay only for a majority of tasks or for the weak tasks; Borda and the Condorcet requirement lie between. Reach is a different matter: on OLLB the median has a lower overall feasible share and a shorter all-feasible prefix than the mean while reaching isolated targets farther down, the minimum is cheap when feasible but needs every weak task raised and is limited by the cap, and on HELM the point estimates for the median, Condorcet-type rules and plurality exceed the mean's, with overlapping intervals for median and mean.

Conditions under which the conclusions fail or are not established. (1) *The cap.* Feasibility is determined by the cap, not by the rule, far from the top (oracle rows of Appendix Table 3), and the ordering of the rules by feasible share changes with the shape of the cap (per-task empirical against uniform). The OLLB cap uses gains of published fine-tunes over their declared base models; this is selection-biased toward successes and rests on about five tail points per task; a bootstrap over the pairs gives wide, upward-skewed intervals (OLLB mean share 11–17%, HELM 15–42%; Appendix Table 7) that cover sampling noise only, not this bias, and was not run for the rank rules. The HELM cap is a transfer of the OLLB MMLU-PRO gain to 57 subjects and has not been measured on HELM. (2) *Independence across tasks.* Each task is raised at no cost to others; real trade-offs would raise the cost. (3) *Fixed rivals.* Others do not respond; strategic rivals need a game model such as Chen et al. [2026]. (4) *Points, not effort.* "Cheap" means few score points, not little compute or data. (5) *Ranks within noise.* The score margin of 10^{-3} points (the witness granularity for rank rules) that defines "first" is below the evaluation noise of the scores. The noise runs show that the optimal requirement near the top varies by multiples of its median; the fixed-policy check shows that a win by the smallest tested margin (score rules) or witness (rank rules) survives resampling in a minority of draws for every rule except the mean, where it is a coin flip (Section 5). (6) *Rule coverage and size.* Rank rules were solved (under the strict model, Appendix C) for the 49 best non-leading OLLB targets only (and on the 500-best slice), so their feasibility fractions there cannot be compared with those of the score rules over all targets; Copeland was solved on HELM and the 500-best slice only, and the Condorcet requirement upper-bounds it. (7) *Small n and M.* OLLB has six tasks and HELM 78 targets, of which at most 15 are feasible under the mean at the primary cap, so HELM rank bins hold few targets and their intervals are wide; scaling in n comes from HELM subsets only (own-rule targets, and a common-target control with 60 draws per n) and is a statement about random subsets of one benchmark family; both datasets are snapshots. (8) *Not done.* Kemeny, Schulze, the trimmed mean (a solver exists, no run), Copeland beyond the 500-best OLLB slice, a bootstrap of the cap for the rank rules, and survival under noise correlated across models.

Broader impact. The analysis shows leaderboard designers how exposed each rule is to targeted training. It could be read as a recipe for gaming a ranking; we think the risk is small, as first place is infeasible for most models under observed gains.

Reproducibility. All numbers, tables and figures come from the supplementary code on pinned public data; versions, checksums, seeds, commands, run times (CPU only) and the script-to-figure/table map are in `REPRODUCE.md` and `figures/MANIFEST.md`. Only derived tables are released; no forbidden material was searched for or met (`README.md`).

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A Proofs of Propositions 1–3

Proposition 1. $f(y) \geq b$ iff $\sum_t (y_t - x_t) \geq R$. Each raise $y_t - x_t$ lies in $[0, h_t]$, so the total can reach R iff $\sum_t h_t \geq R$, and the minimum of the (linear) total is R, attained by any feasible spread. Touching k tasks supplies at most the sum of the k largest h_t, which gives K. A raise with maximum d supplies at most $\sum_t \min(d, h_t)$, so L_∞ is the least d with $\sum_t \min(d, h_t) \geq R$; at that d the capped uniform raise supplies exactly R (the sum is continuous and non-decreasing in d), so its total cost is R as well. Without a binding cap ($d \leq \min_t h_t$) $\sum_t \min(d, h_t) = nd$ and $d = R/n$.

Proposition 2. $\min_t y_t \geq b$ requires $y_t \geq b$ for every t, hence $c_t \geq b$ wherever $x_t < b$ (tasks with $x_t \geq b$ need no change). The cheapest y_t on a deficient task is b, so $L_1 = \sum_{t \in D} (b - x_t)$, $K = |D|$, and L_∞ is the largest of these raises. Each is forced.

Proposition 3. For odd n the median is at least b iff at least m coordinates are. Tasks with $x_t \geq b$ already count, so $m' = (m - a)^+$ more are needed. Lifting task t to b costs d_t and is possible iff $c_t \geq b$; costs are separable, so the cheapest set is the m' smallest finite d_t and $K = m'$, which cannot be smaller because each lifted task adds one to the count.

B Further results

Table 2: Closed forms (Propositions 1–3) against the exact solvers and an independent generic LP/MILP (60-target sample), over all caps. Generated by `figures/src/tables.py` from `results/theory_check.csv`.

data	rule	reference	max$\lvert\Delta L_1\rvert$	max$\lvert\Delta K\rvert$	feas. mismatches	caps
OLLB	mean	exact solver	5.7e-14	0	0	7
OLLB	mean	generic LP/MILP	8.5e-14	0	0	7
OLLB	min	exact solver	0.0e+00	0	0	7
OLLB	min	generic LP/MILP	4.3e-14	0	0	7
OLLB	median	exact solver	solver only	–	–	7
HELM	mean	exact solver	4.5e-13	0	0	8
HELM	mean	generic LP/MILP	1.8e-12	0	0	8
HELM	min	exact solver	0.0e+00	0	0	8
HELM	min	generic LP/MILP	1.3e-12	0	0	8
HELM	median	exact solver	5.7e-14	0	0	8
HELM	median	generic LP/MILP	1.5e-12	0	0	8

Table 3: Feasibility frontier under the primary cap (`emp99` on OLLB, `emp99T` on HELM) and the oracle cap (`ceil`). *share*: fraction of targets for which first place is reachable (Wilson 95% intervals); r^*: largest rank under the mean among feasible targets (may be an isolated case); *all feas. up to*: largest r such that every target of rank $\leq r$ is feasible; *first bin < 1/2*: first rank bin (bins double from rank 2) in which fewer than half are feasible. †: the 49 non-leading targets among the 50 best under the mean only, not comparable with the score rules' fractions. Generated by `figures/src/tables.py`.

cap	rule	targets	feasible	share [95% CI]	r^*	all feas. up to	first bin < 1/2
primary (`emp99`)	mean	4495	546	0.12 [0.11, 0.13]	901	394	513–1024
primary (`emp99`)	median	4495	451	0.10 [0.09, 0.11]	1207	90	513–1024
primary (`emp99`)	min	4495	273	0.06 [0.05, 0.07]	2225	5	65–128
primary (`emp99`)	Borda†	49	49	1.00 [0.93, 1.00]	50	50	none
primary (`emp99`)	Condorcet†	49	42	0.86 [0.73, 0.93]	50	18	none
oracle (`ceil`)	mean	4495	4495	1.00 [1.00, 1.00]	4496	4496	none
oracle (`ceil`)	median	4495	4495	1.00 [1.00, 1.00]	4496	4496	none
oracle (`ceil`)	min	4495	4495	1.00 [1.00, 1.00]	4496	4496	none
oracle (`ceil`)	Borda†	49	49	1.00 [0.93, 1.00]	50	50	none
oracle (`ceil`)	Condorcet†	49	49	1.00 [0.93, 1.00]	50	50	none

cap	rule	targets	feasible	share [95% CI]	r^*	all feas. up to	first bin < 1/2
primary (`emp99T`)	mean	78	15	0.19 [0.12, 0.29]	16	16	17–32
primary (`emp99T`)	median	78	26	0.33 [0.24, 0.44]	29	19	33–64
primary (`emp99T`)	min	78	9	0.12 [0.06, 0.20]	20	6	9–16
primary (`emp99T`)	Borda	78	16	0.21 [0.13, 0.31]	17	17	17–32
primary (`emp99T`)	Condorcet	78	25	0.32 [0.23, 0.43]	26	26	33–64
oracle (`ceil`)	mean	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	median	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	min	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	Borda	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	Condorcet	78	78	1.00 [0.95, 1.00]	79	79	none

Variants and sensitivity checks

All numbers below are in `results/summary/summary_variants.csv` (script `code/summarize_variants.py`). Ratios are over targets feasible both in the variant and in the core run.

Table 4: The four non-leading targets of mean-rank 2–5: minimum L_1 in points (K in parentheses) under each rule; OLLB (`emp99`, upper) and HELM (`emp99T`, lower). Bold: cheapest rule(s) for that target; "0.00 (1)" is a raise of the 0.001 margin, "0.00 (0)" means the target is already first under that rule. Last row: number of the four targets for which the rule is strictly cheaper than the mean. With four targets per table these are exceptions to a general ordering, not rates. Generated by `figures/src/tables_v2.py`.

target (mean-rank)	mean	median	min	Borda	Condorcet	plurality
MaziyarPanahi/calme-3.1-instruct-78b (2)	4.77 (1)	9.55 (1)	3.68 (1)	**1.65 (1)**	6.70 (3)	3.50 (2)
dfurman/CalmeRys-78B-Orpo-v0.1 (3)	5.11 (1)	8.68 (1)	3.12 (1)	**0.00 (0)**	5.29 (2)	5.55 (2)
MaziyarPanahi/calme-2.4-rys-78b (4)	7.90 (1)	8.37 (1)	2.79 (1)	**0.74 (1)**	6.25 (2)	7.47 (2)
huihui-ai/Qwen2.5-72B-Instruct-abliterated (5)	23.86 (1)	**0.70 (1)**	14.60 (2)	6.23 (1)	9.72 (3)	15.38 (2)
cheaper than mean, of 4	–	1	4	4	2	3

target (mean-rank)	mean	median	min	Borda	Condorcet	Copeland	plurality
DeepSeek v3 (2)	4.04 (1)	**0.00 (1)**	**0.00 (0)**	5.58 (5)	**0.00 (1)**	**0.00 (1)**	1.42 (4)
Gemini 1.5 Pro (002) (3)	22.53 (4)	3.25 (4)	3.01 (1)	21.10 (10)	4.64 (8)	4.63 (8)	**0.00 (2)**
Claude 3.5 Sonnet (20240620) (4)	41.80 (6)	1.53 (3)	2.28 (2)	0.46 (3)	0.89 (7)	0.89 (7)	**0.00 (0)**
Claude 3 Opus (20240229) (5)	154.04 (22)	6.22 (5)	7.57 (3)	43.67 (20)	24.53 (18)	24.53 (18)	**5.04 (7)**
cheaper than mean, of 4	–	4	4	3	4	4	4

Margin. With margins of 1 and 2 points instead of 10^{-3} the feasible share on OLLB falls from 12.1% to 11.1% and 10.4% (mean), from 10.0% to 9.0% and 7.6% (median) and from 6.1% to 4.3% and 2.6% (minimum); the median ratio of the cost to the minimal-margin cost is 1.08 and 1.17 (mean), 1.15 and 1.34 (median), 1.25 and 1.53 (minimum). On HELM the shares fall from 19.2% to 15.4% and 11.5% (mean), from 33.3% to 28.2% and 20.5% (median), and from 11.5% to 11.5% and 9.0% (minimum), with ratios of 1.3 and 1.7 (mean), 1.5 and 2.3 (median). Margin variants exist for the score rules only: the rank rules use strict comparisons without a margin (Appendix C). The ordering of the feasible shares is unchanged on OLLB.

Slice of the 500 best models. To extend the rank rules beyond the 50 best targets we solved them on a leaderboard that keeps only the 500 best models by the mean (499 non-leading targets; time-outs are counted as censored, not as infeasible, and none remain under `emp99` after a 900 s re-solve). Under `emp99` the feasible shares are 94.8% (mean), 90.6% (Borda), 79.4% (median), 74.9% (Copeland), 56.3% (plurality), 40.3% (minimum) and 30.5% (Condorcet requirement, which is sufficient for Copeland and hence below it); with the oracle cap every target is feasible under every rule (Borda: 99.8% with one censored instance). These shares refer to a different set of rivals than the full leaderboard, and the rules are again not compared on a common footing beyond what the paired columns of `summary_variants.csv` show.

Merged models, scale and weights. Dropping the merged models (3788 targets) leaves the OLLB shares nearly unchanged (mean 11.5%, median 9.4%, minimum 7.0%). With raw task accuracies instead of the leaderboard's normalised scores the ordering of the feasible shares under a +10 point cap changes (normalised: minimum 9.9%, median 4.5%, mean 2.5%; raw: mean 6.6%, minimum 5.0%, median 0.5%), so cap comparisons in points depend on the score scale; we use the leaderboard's own normalisation. Random task weights drawn from a Dirichlet distribution (concentration 1 or 5; the stratified targets of ranks 2, 3, 5, 10, 20, 50, 100 on OLLB and 2, 3, 5, 10, 20, 40 on HELM, 500 draws each) give a median weighted cost of 0.54 and 0.70 (OLLB) and 0.43 and 0.68 (HELM) times the equal-weight cost of the same target, so weights that concentrate on tasks where the target is strong make first place cheaper.

Table 5: Capped uniform raise of every task by the same amount against the optimal raise, primary cap (OLLB `emp99`, upper; HELM `emp99T`, lower): ratio $L_1^{\text{unif}}/L_1^{\text{opt}}$, median over targets feasible under both with positive cost [95% bootstrap interval over targets]; *ratio* < 1: targets with a ratio below 1 (rank rules only, at most $n\varepsilon/L_1^{\text{opt}}$ because the optimum is a witness cost); K: medians. †: OLLB rank rules cover the 50 best targets only. The sets of feasible targets differ by rule. Generated by `figures/src/tables_v3.py`.

rule	feas. opt/unif	n_{both}	$L_1^{\text{unif}}/L_1^{\text{opt}}$ [95% CI]	ratio < 1	K^{opt}	K^{unif}
OLLB, primary cap (`emp99`)						
mean	546/546	546	1.00 [1.00, 1.00]	0	4	6
median	451/451	450	2.38 [2.36, 2.42]	0	3	6
min	273/273	272	3.64 [3.58, 3.74]	0	2	6
Borda†	49/49	48	2.61 [2.27, 2.71]	0	2	6
Condorcet†	42/42	42	2.39 [2.13, 2.72]	0	4	6
plurality†	36/36	36	3.00 [2.64, 3.24]	0	2	6
OLLB, oracle cap (`ceil`)						
mean	4495/4495	4495	1.00 [1.00, 1.00]	0	2	6
median	4495/4495	4494	2.01 [2.00, 2.02]	0	3	6
min	4495/4495	4494	2.41 [2.39, 2.44]	0	3	6
Borda†	49/49	48	2.62 [2.30, 2.72]	0	2	6
Condorcet†	49/49	49	2.41 [2.23, 2.80]	0	3	6
plurality†	49/49	49	3.47 [3.31, 3.81]	0	2	6

rule	feas. opt/unif	n_{both}	$L_1^{\text{unif}}/L_1^{\text{opt}}$ [95% CI]	ratio < 1	K^{opt}	K^{unif}
HELM, primary cap (`emp99T`)						
mean	15/15	15	1.00 [1.00, 1.00]	0	32	57
median	26/26	26	8.96 [8.41, 11.31]	0	10	57
min	9/9	8	37.54 [28.51, 53.60]	0	2	57
Borda	16/16	16	1.85 [1.58, 2.30]	0	33	57
Condorcet	25/25	25	2.99 [2.89, 3.20]	1	27	57
plurality	45/45	44	8.00 [7.60, 8.74]	1	9	57
Copeland	25/25	25	2.99 [2.89, 3.20]	1	27	57
HELM, oracle cap (`ceil`)						
mean	78/78	78	1.00 [1.00, 1.00]	0	16	57
median	78/78	78	5.12 [4.17, 5.83]	0	21	57
min	78/78	77	9.49 [7.00, 11.54]	0	12	57
Borda	78/78	78	1.40 [1.35, 1.45]	0	47	57
Condorcet	78/78	78	2.70 [2.63, 2.77]	1	29	57
plurality	78/78	77	7.64 [7.43, 7.78]	1	9	57
Copeland	78/78	78	2.70 [2.64, 2.77]	1	29	57

Further score rules. Geometric, harmonic and squared power means and plurality were solved on the same targets; their results are in `results/core_*.csv` and the frontier tables `results/summary/frontier_*.csv`, and we make no separate claim about them in the text. A trimmed mean has a solver in the code but was not run.

C Exactness of the rank-rule solver

The rank rules (Borda, Condorcet requirement, plurality, Copeland) depend only on the order of scores on each task. We therefore use one model for every cost and for the winner test. A pairwise comparison of two scores is a win (strictly higher), a loss, or a tie worth one half; Borda adds, per task, the models strictly below plus half the ties; the Condorcet requirement asks for strictly more than $n/2$ pairwise points against every rival; Copeland scores each rival beaten as 1 and tied as one half; plurality gives each task one point split equally among all m models tied for its highest score (so a tie is worth $1/m$, not one half). Under Borda, plurality and Copeland the target is first if its total is strictly the largest; under the Condorcet requirement it is first if it has more than $n/2$ pairwise points against each rival. There is no margin. A strict win is determined by order, so the strict-winning infimum of L_1 need not be attained; it is attained when a tie suffices (for Borda, a target with scores $(0, 1)$ against rivals $(1, 0)$ and $(1, 0.5)$, cap $(1, 1)$: any task-0 score below 1 leaves it behind the second rival, a score of 1 ties both rivals on that task and wins, so the minimum $L_1 = 1$ is attained at a tie). The solver restricts each raised task to a rival's score (a tie) or to $\min(s + \varepsilon, c_t)$ with $c_t > s$ (strictly above a rival score s, at most ε above it), and returns the cost of that explicit witness. Two consequences follow. Feasibility and K do not depend on ε and are those of continuous raises; L_∞ is

Table 6: Survival of the optimised raise under evaluation noise, the raise fixed (not re-optimised): share of noise draws in which the target is still strictly first; median over targets, min–max over targets, and the median with no raise. ε is the score margin (0.001, the smallest used, 1 or 2 points) for the mean, median and minimum, and the witness granularity (0.001) for the rank rules. OLLB: 7 targets of mean-rank 2–100, 300 draws each; HELM: 6 targets of rank 2–40, 200 draws each; infeasible and already-first targets are excluded, so the target count falls under some rules. Medians over 2–7 targets are descriptive. Generated by `figures/src/tables_v3.py`.

rule	ε (pts)	targets	median survival	min–max	median, no raise
OLLB, cap `emp99`, 300 noise draws per target					
mean	0.001	7	0.43	0.40–0.50	0.00
mean	1	7	0.80	0.79–0.87	0.00
mean	2	7	0.98	0.96–0.99	0.00
median	0.001	7	0.21	0.10–0.28	0.00
median	1	7	0.72	0.59–0.83	0.00
median	2	7	0.96	0.95–0.99	0.00
min	0.001	4	0.11	0.04–0.17	0.00
min	1	4	0.21	0.14–0.29	0.00
min	2	2	0.45	0.45–0.45	0.00
Borda	0.001	6	0.19	0.08–0.34	0.01
Condorcet	0.001	6	0.01	0.00–0.07	0.00
plurality	0.001	5	0.18	0.12–0.20	0.00
HELM, cap `emp99T`, 200 noise draws per target					
mean	0.001	4	0.38	0.37–0.49	0.04
mean	1	4	0.96	0.95–0.98	0.04
mean	2	4	1.00	1.00–1.00	0.04
median	0.001	4	0.05	0.00–0.19	0.02
median	1	4	0.11	0.01–0.27	0.02
median	2	4	0.28	0.07–0.41	0.02
min	0.001	3	0.05	0.04–0.31	0.01
min	1	3	0.07	0.07–0.40	0.01
min	2	3	0.11	0.09–0.47	0.01
Borda	0.001	4	0.05	0.03–0.16	0.00
Condorcet	0.001	5	0.00	0.00–0.17	0.00
plurality	0.001	5	0.00	0.00–0.18	0.00

Table 7: Cap bootstrap (1000 resamples of the 540 fine-tune/base pairs, each pair drawn as a whole row with all its tasks so the dependence across tasks is kept; HELM: the transferred OLLB MMLU-PRO gains): feasible share, largest feasible rank under the mean and all-feasible prefix, observed value with the 2.5–97.5 percentile interval. The intervals cover sampling noise of the cap only, not the selection bias of published fine-tunes or the HELM transfer; rank rules were not bootstrapped. Generated by `figures/src/tables_v3.py`.

cap	rule	feasible share [95% CI]	largest feasible rank [CI]	all feasible up to [CI]
OLLB, bootstrap over fine-tune/base pairs (B = 1000)				
`emp99`	mean	0.121 [0.112, 0.166]	901 [824, 1201]	394 [346, 590]
`emp99`	median	0.100 [0.083, 0.121]	1207 [1207, 1634]	90 [70, 216]
`emp99`	min	0.061 [0.050, 0.082]	2225 [1464, 2225]	5 [4, 6]
`emp95`	mean	0.075 [0.065, 0.089]	447 [411, 566]	109 [85, 138]
`emp95`	median	0.031 [0.026, 0.040]	715 [707, 771]	12 [12, 12]
`emp95`	min	0.014 [0.010, 0.025]	1407 [1329, 1407]	4 [1, 4]
HELM, bootstrap over fine-tune/base pairs (B = 1000)				
`emp99T`	mean	0.192 [0.154, 0.423]	16 [13, 34]	16 [13, 34]
`emp99T`	median	0.333 [0.205, 0.551]	29 [22, 46]	19 [13, 40]
`emp99T`	min	0.115 [0.090, 0.205]	20 [20, 34]	6 [6, 6]
`emp95T`	mean	0.090 [0.038, 0.103]	8 [4, 9]	8 [4, 9]
`emp95T`	median	0.128 [0.077, 0.167]	11 [8, 21]	11 [5, 12]
`emp95T`	min	0.064 [0.051, 0.077]	9 [9, 20]	4 [2, 4]

the infimum itself (bisection, tolerance 10^{-6}). The infimum of L_1 lies in $[L_1 - n\varepsilon, L_1]$ (n tasks): by monotonicity, any winning continuous raise can be replaced by an allowed witness that costs at most ε more per raised task, so the witness optimum is at most the infimum plus $n\varepsilon$, and it is never below the infimum. The bound uses $n\varepsilon$ and not $K\varepsilon$, because the K reported beside L_1 is minimised separately and need not be the number of tasks raised in the witness. An earlier version used finite levels at $s + \varepsilon$ credited with passing every lower rival; a review of the code showed that this model could disagree with L_∞ and with the pairwise-margin definition on closely spaced scores (a level generated from one rival could pass another by less than ε and be credited in full), and the paper's earlier rank-rule

costs came from it. The strict model replaced it, with plurality solved on an exact credit lattice and an exact rational check of every witness. The unit tests compare the solver with brute force on fractional, tied, sub-ε and cap-boundary instances and on the reviewers' counterexamples. We regenerated every rank-rule result with it (`results/audit_ordinal_fix.csv`): no feasibility verdict changed in any file, only Borda costs changed in the primary files (4 of 100 rows of the OLLB top-50 file, largest change 0.10 points; none on HELM), and more in the sensitivity runs (Borda on the 500-best slice 116 of 1000 rows, up to 1.35 points; HELM cap sweep 9 of 395 rows, up to 3.5 points; OLLB task bootstrap 20 of 420 rows, up to 1.6 points). All numbers in this paper come from the regenerated files. For score rules a target already ahead of every rival costs zero, whatever the size of its lead; this matters for the margin variants only for the leaders of the median and minimum rules among the mean-nonleaders. L_∞ is not reported for rank rules in the main text, and the rank-rule noise and bootstrap rows are over the same strict model.

D Hypotheses fixed before the full results

The plan fixed three hypotheses from the closed forms before the full result set was available. H1: with loose caps, score-sum rules need the same L_1 regardless of K, while order-statistic and majority rules need more tasks but possibly fewer points than the mean. H2: required investment grows with the target's initial rank; feasibility falls as caps tighten; rule ordering can change with the cap. H3: for fixed per-task caps the share K/n of tasks that majority and median rules need does not shrink with n, while the mean rule's L_1/n depends on the density of gaps. Anything beyond these is exploratory.

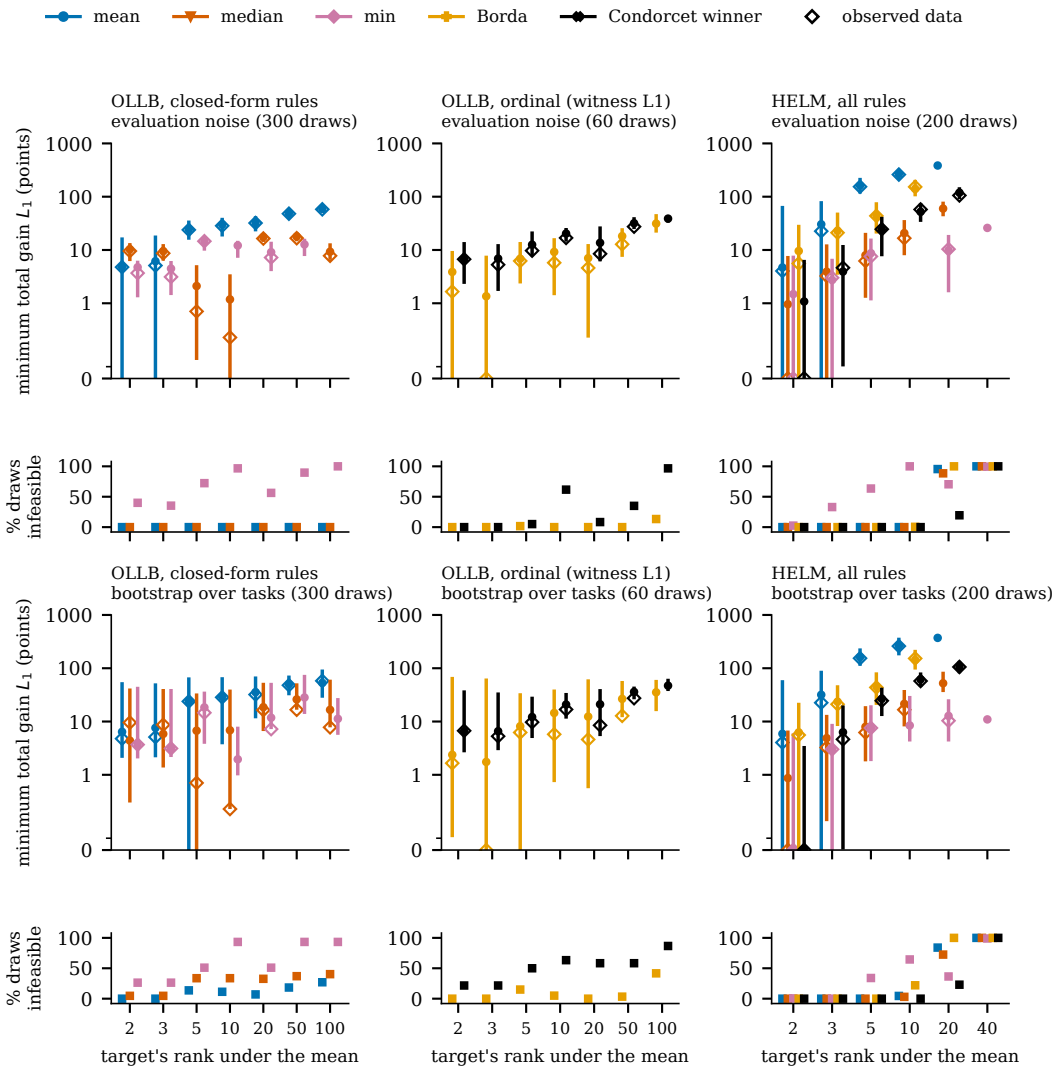


Figure 2: Stability of the optimal requirement (L_1) under evaluation noise (top row) and a bootstrap over tasks (bottom row), for target ranks 2, 3, 5, 10, 20, 50, 100 (OLLB, `emp99`) and 2, 3, 5, 10, 20, 40 (HELM, `emp99T`), per rule. Dot: median over feasible draws; bar: 2.5–97.5 percentile of those draws (a range of draws, not a confidence interval of a median); open diamond: value on the observed data (absent where the target is outside the solved ordinal set or infeasible; rank-rule L_1 is the witness cost, within $n\varepsilon$ of the strict infimum); lower strip: percentage of draws in which first place is unreachable. Noise: independent Gaussian draws with the binomial standard error of each score (300 draws for OLLB closed forms, 60 for OLLB rank rules, 200 for HELM; the task bootstrap uses the same counts; OLLB item counts come from one results file and are assumed equal for all models). Each draw is re-optimised, so the figure shows variability of the *optimal requirement*, not whether a fixed raise remains a win. At ranks 2–3 the lower bar end is 0 because some draws already make the target first. Counted for the evaluation-noise row (feasible target-by-rule cases of the five rules shown), the observed cost lies inside the 2.5–97.5 range in every case (OLLB closed forms 18/18, OLLB rank rules 12/12, HELM 21/21); with ranges this wide this does not bound a shift. Generated by `figures/src/fig_robust.py`.

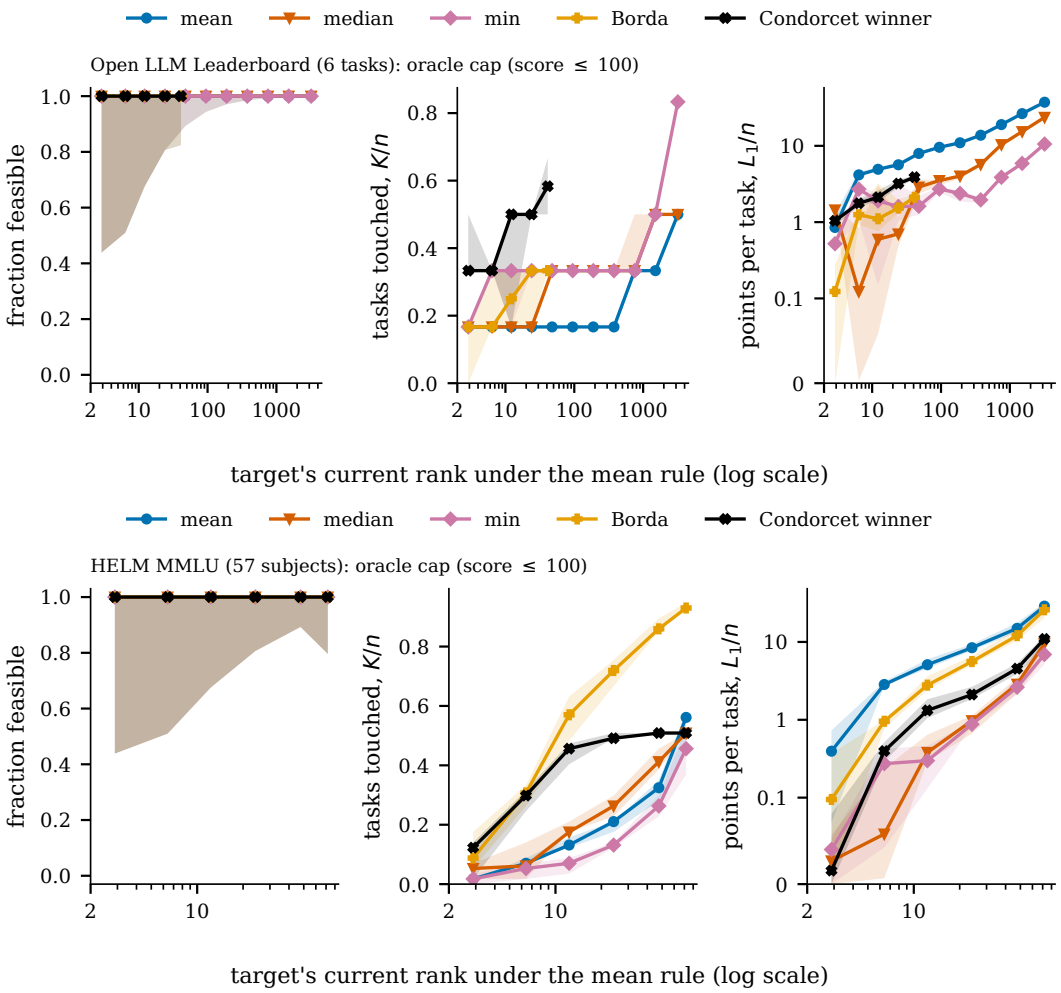
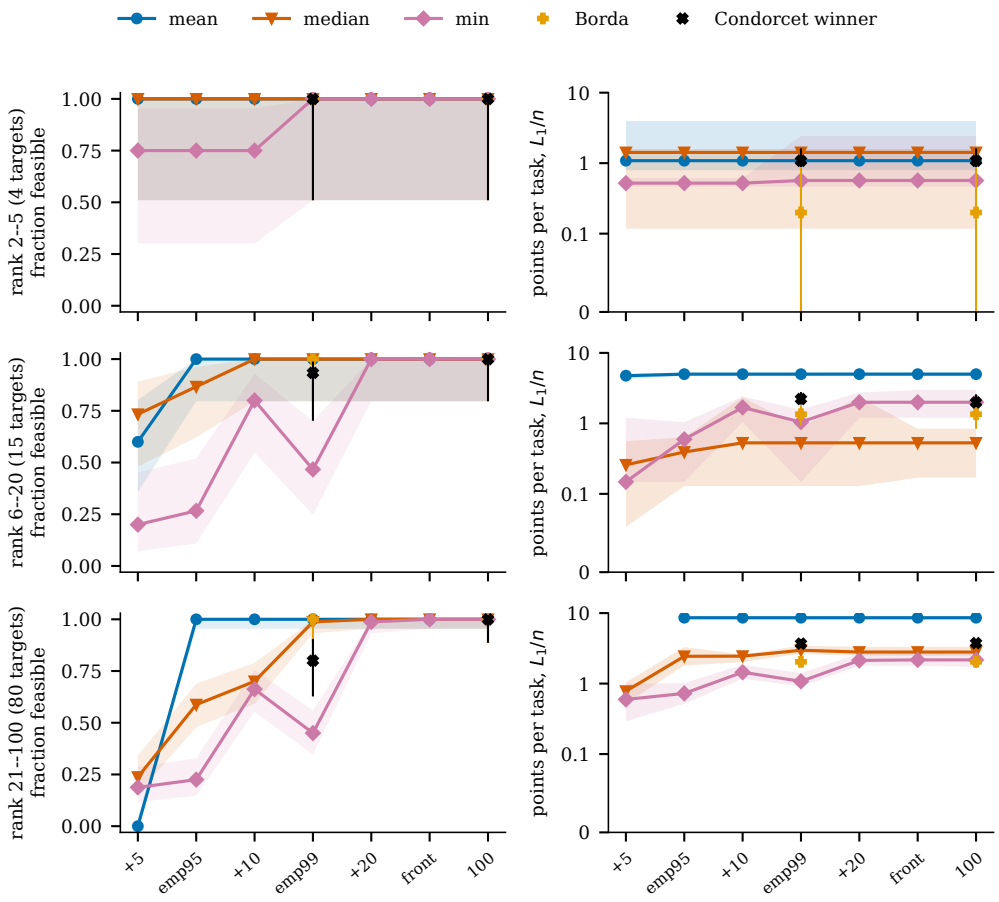
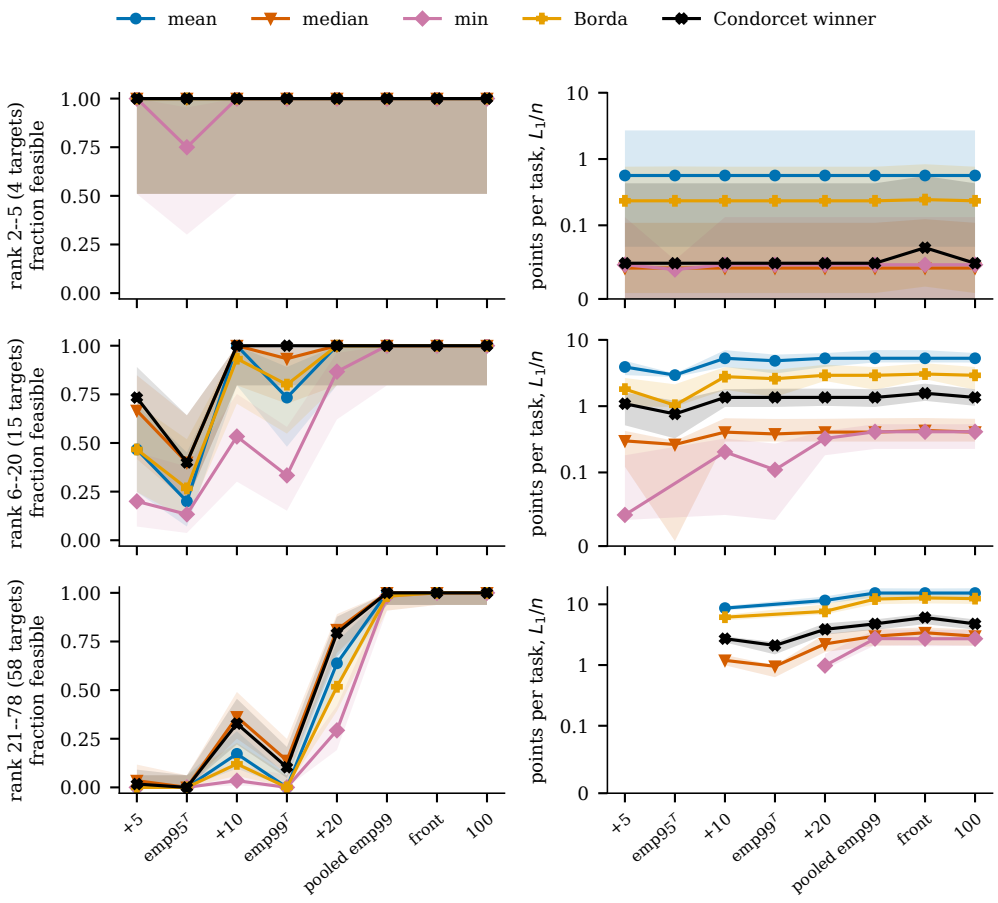


Figure 3: As Figure 1 with the oracle cap `ceil` (scores up to 100): a lower bound on any required investment; every target is feasible, so the differences between rules are not caused by the cap.



cap scenario (not nested, no monotone order; +h: at most h points on every task; emp: per-task empirical fine-tune gain percentile; ^{T}: transferred from OLLB; front: best rival score on each task; 100: score ceiling)

Figure 4: OLLB: feasibility (left) and median L_1/n among feasible targets (right) for three rank strata under the mean (mean-ranks 2–5, 6–20, 21–100; $N = 4$ in the first, so its intervals are wide), under each cap *scenario*. The horizontal axis orders scenarios and is not a monotone sweep: $+h$ is a uniform cap, `emp95` and `emp99` are per-task percentile caps, "front" is the best rival score on each task, 100 is the ceiling; these cap vectors are not nested. Two things move the right panels: the feasible cohort changes with the cap, so a median over feasible targets changes even if no target's own optimum does; and a target's own optimum, which under the mean (and for the minimum, when feasible) is the same under any cap under which it is feasible (Proposition 1), while for the median and rank rules no such invariance is established and relaxing a cap can allow a cheaper allocation. Where the mean line is flat the cohort did not change; where it moves (6–20 at $+5$) it did. Bands: Wilson 95% (left), bootstrap over targets (right), absent with fewer than 3 feasible targets. Borda and Condorcet (50 best targets only) are unconnected markers and not comparable with the score rules' fractions in the 21–100 stratum. Generated by `figures/src/fig_cap_grid.py`.



cap scenario (not nested, no monotone order; +h: at most h points on every task; emp: per-task empirical fine-tune gain percentile; ^{T}: transferred from OLLB; front: best rival score on each task; 100: score ceiling)

Figure 5: HELM: as Figure 4 (strata 2–5, 6–20, 21–78); T denotes the transfer of the OLLB MMLU-PRO gains to all 57 subjects, so the scenarios mix a transfer assumption with the cap shape.

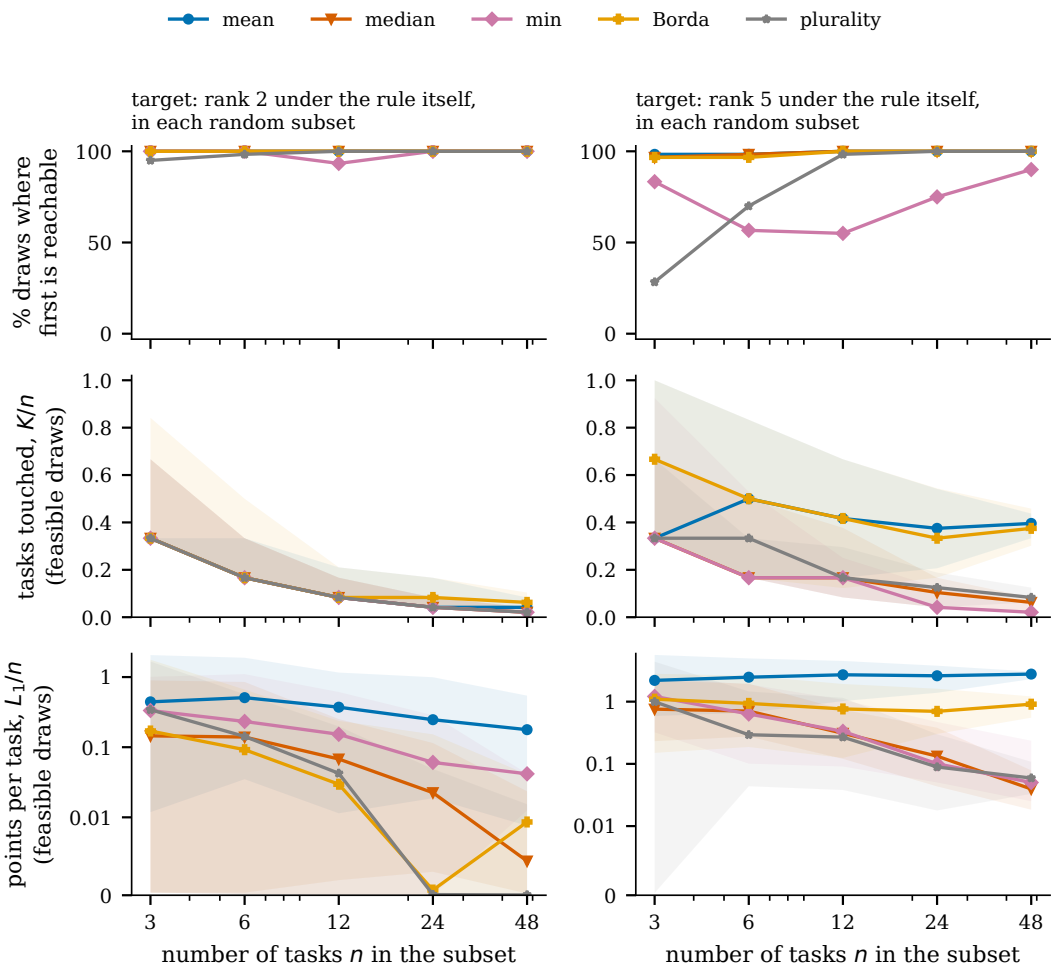


Figure 6: HELM MMLU: random subsets of n of the 57 subjects (60 subsets per n), cap `emp99T`. In every subset and for every rule separately, the target is the model ranked 2 (left) or 5 (right) *under that rule on that subset*, so the target differs between rules and subsets and the curves mix the effect of the rule with that of the target; it is not the mean-ranked model. Top row: percentage of subsets in which first place is reachable within the cap (counts in `F_helm_scaling.json`). Middle and bottom rows: K/n and L_1/n, medians over the *feasible* subsets only; bands are the 2.5–97.5 percentile range of those subsets (a range of draws, not a confidence interval of the median). Where the top row is below 100% the cost curve is conditional on a selected set of subsets. The Condorcet requirement is not in this run. Generated by `figures/src/fig_scaling.py`.

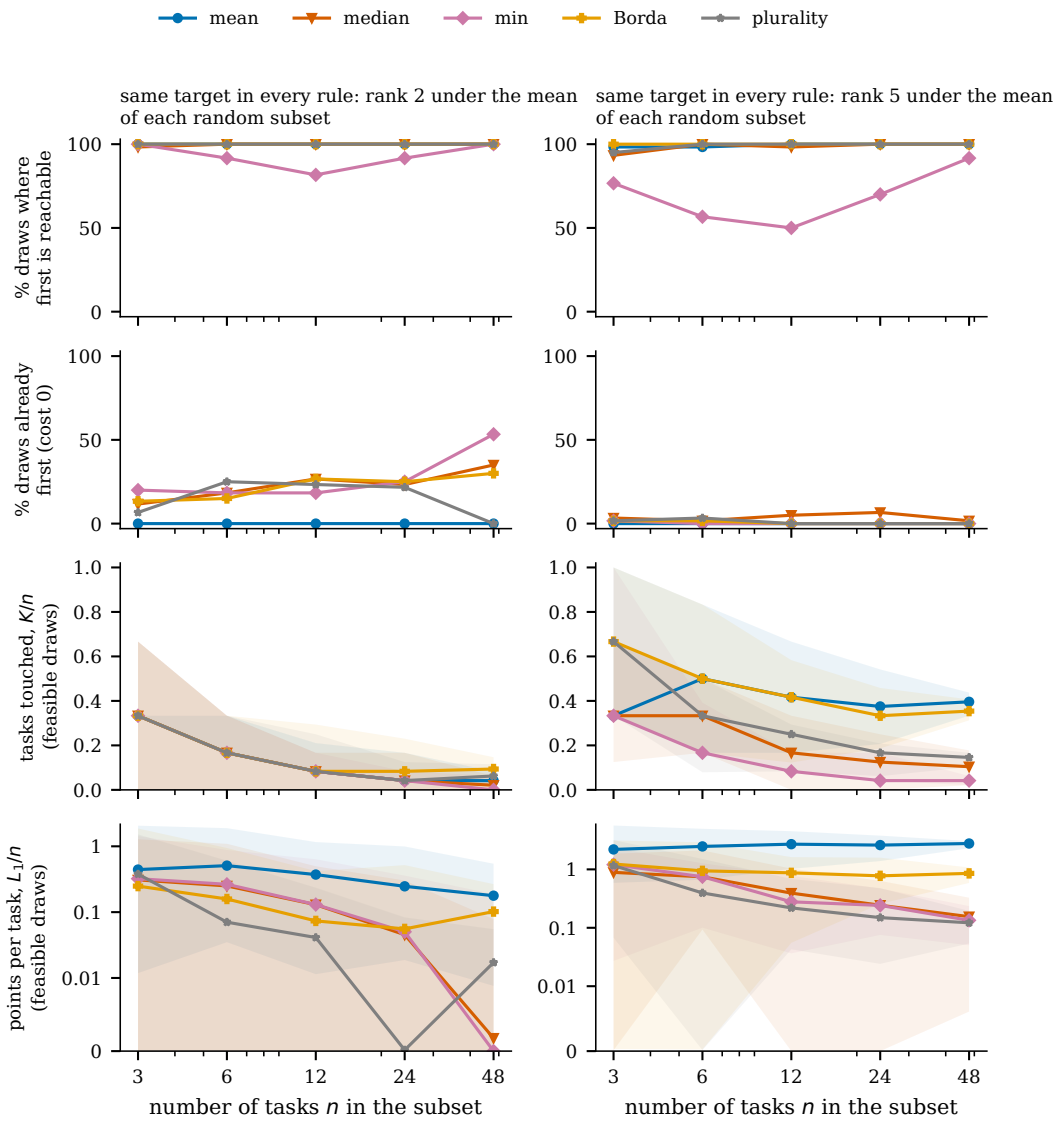


Figure 7: HELM scaling control: as Figure 6, but the target is the same model under every rule within a draw (rank 2 or 5 under the *mean* of the subset; 60 draws per n, cap `emp99T`). Rows: share of draws in which first place is reachable; share in which the target is already first (cost 0, included in the cost rows); median K/n and L_1/n over feasible draws with the 2.5–97.5 percentile range of those draws (a spread of draws, not an interval for the median); the L_1/n axis is symlog. Only this version supports a statement about how a rule reacts to n. Generated by `figures/src/fig_scaling_common.py`.

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   Justification: Bootstrap intervals over targets (Figure 1, Table 1), Wilson intervals for feasible shares (Appendix Table 3), a bootstrap over the gain pairs for the cap percentile (feasible shares and frontier ranks, Section 5), Monte Carlo ranges for evaluation noise and task resampling on both datasets (Figure 2), and the share of draws in which a fixed raise still wins, stated as ranges or medians over draws. Not quantified: the cap bootstrap for rank rules and noise correlated across models (Section 6).

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